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1.1 WHAT IS LIFE ?

‘**Life**’ is an exclusive property of the various types of living organisms in the world. Recent discovery of fossils of some microscopic organisms extended the known ‘history of life’ on the Earth to about **3.5** billion years. **Biology** is the science dealing with the study of living organisms which are broadly classified into microorganisms, plants and animals. In spite of their wide ‘**diversity**’, they show ‘**fundamental uniformity**’ in possessing *protoplasm* with the genetic substance, the *DNA* which is made up of ‘**nucleotides**’, universally. Living organisms show:

- I. **Cellular organization** and **highly ordered structure**: An organism’s body is made up of one to many ‘**cells**’, which are the **structural** and **functional units** of the body. Each cell is isolated from the cells surrounding it by a limiting membrane the plasma membrane. This membrane controls the exchange of various substances across it. Two major kinds of cells occur in the living world – the **prokaryotic** cells (without distinct nuclei) and the **eukaryotic** cells (with distinct nuclei).
- II. **Complexity of organization**: A fundamental feature of life is the presence of a high degree of ‘order’. Living organisms show hierarchical organization such as cellular level, tissue level, organ level and organ- system level. Several organ-systems make up an **organism**. This type of ‘**ordered complexity**’ is not seen in the non-living things. The hierarchy of life can be schematically represented as follows (starting from the inorganic elements that go into the constitution of protoplasm).



ATOMS → MOLECULES → CELL → TISSUE → ORGAN → ORGAN SYSTEM → ORGANISM → POPULATION → COMMUNITY → ECOSYSTEM → BIOSPHERE.

This representation includes both the inorganic and organic components, as living substance is made up of several inorganic constituents and life processes involve continuous exchange and recycling of inorganic elements.

- III. Sensitivity/ Response to Environment:** It is the property of showing response to external or internal stimuli received through various kinds of 'sense organs'. Living organisms show response to environmental stimuli, which could be physical, chemical or biological. Plants show sensitivity to environmental factors such as light, temperature, water, etc. Ability to show response is called irritability
- IV. Growth:** Growth is one of the *fundamental characters* of living beings (including the unicellular organisms). Increase in mass and increase in number are twin characteristics of growth. Some non-living things also may show tendency of growth but growth in living beings is '**growth from inside**', whereas growth in the non-living things is by accumulation of material *on the surface*. Higher plants show growth all their life producing new branches, leaves, etc., where as it is

'limited' (up to a certain age only) in animals. *Growth cannot be taken as a defining property of living organisms (Ref: NCERT Text Book) as non-living structures also show growth in some respects.*

- V. Energy utilisation/ Energy processing/ Metabolism:** The sum total of all the chemical reactions occurring in the bodies of organisms constitute metabolism and it is a defining feature of all living organisms without exception. Life activities require 'energy' in different forms. The chief source of energy for living organisms is the *sun light*. Energy is transferred from one trophic level to another level. The life processes which build up or store (*conserve*) energy are called **anabolic** processes and the processes involving expenditure of energy are called **catabolic** processes. Together, they constitute - **metabolism**.

ATP is the **chief energy carrier** for various reactions in living systems and it is appropriately described as the 'Cellular Energy Currency'.

- VI. Reproduction:** Living organisms produce young ones of their kind, using molecules of heredity, the DNA molecules (genes), which are passed on to the offspring. It is important to note that '**life comes only from life**' (**biogenesis**) and not from non-living substances. Reproduction, characteristic of all living organisms, occurs by *vegetative*, *asexual* and *sexual* methods. Sexual method involves fusion of gametes forming a zygote, which through various stages of development becomes a young organism. However, some organisms do not reproduce (mules, sterile worker honey bees, infertile human couples, etc.,). **Protozoans** reproduce by asexual methods such as 'binary fission'. **Sponges, hydra**, etc., reproduce by 'budding'. In **planarians**, fragmented body (by fission) regenerates the lost parts of the body and becomes a new organism.

- VII. Homeostasis:** Maintenance of relatively constant internal conditions (**steady state**) different from the surrounding environment is called '**homeostasis**'. It is the **dynamic constancy** of the **internal environment** of an organism within a range that the cells can tolerate. Living organisms maintain 'constant internal environment' by various physiological adaptations.

- VIII. Evolution:** Life is not 'static'. It constantly changes (*dynamic*) from simple to more complex forms. Variations in organisms arise through '**mutations**' or '**gene re-combinations**'. Charles Darwin's Natural Selection theory states that living beings accumulate their 'beneficial variations' over a period of time and tend to evolve gradually into new types of organisms (species) after surviving the struggle for existence.

- IX. Senescence and Mortality:** Living beings are born, grow into mature forms, undergo ageing process and finally die. The process of ageing is called 'senescence'. It leads to death / mortality. Living beings are thus '**mortal**'.